

Marine Heatwaves: An overlooked Driver of Baltic cyanobacteria blooms?

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Christian-Albrechts-Universität zu Kiel

Dozent:

Dr. rer. nat. John Rapaglia

Authors:

Jonah van den Bos (1155444)

Justin Lingg-Laham (1168916)

Abstract

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English

Marine Heatwave (MHW) events have been growing in frequency and intensity in most regions of the world. The trends of these anomalous events in stratified marginal seas like the Baltic Sea have been studied extensively. At the same time, Harmful Algal Blooms (HABs) represent a recurring ecological phenomenon for this region. While most studies describe increased nutrient input as the main driver of HABs, examples from global study sites have shown that temperature-related anomalies may influence the timing and intensity of bloom development. This study focuses on the relationship between MHW events and HABs in the surface layer of the Baltic Sea using Sentinel-2 data within the cloud-based platform Google Earth Engine (GEE). The goal is to investigate the correlation and influence between MHWs and HABs, identifying time periods where both overlap. For this purpose, cyanobacterial bloom intensity is quantified by calculating a phycocyanin-based index, while MHW events are detected from sea surface temperature (SST) records using a threshold-based detection algorithm. The results show no significant relationship between marine heatwaves and the frequency of cyanobacterial bloom occurrence. However, a significant correlation was observed between MHW duration and bloom intensity. These findings suggest that MHWs act as an amplifying factor for established bloom conditions rather than serving as a primary trigger. These findings contribute to the discussion about whether MHWs could indeed be considered a notable driver of cyanobacteria blooms in the Baltic Sea, backed by already existing studies, research and literature.

Deutsch

Marine Hitzewellen (MHW) treten in den meisten Regionen der Welt sowohl häufiger als auch intensiver auf. Dabei wurden die Trends dieser anomalen Ereignisse in Randmeeren wie der Ostsee umfassend untersucht. Zeitgleich stellen schädliche Algenblüten (HABs) ein wiederkehrendes ökologisches Phänomen für diese Region dar. Während ein Großteil der Untersuchungen einen erhöhten Nährstoffeintrag als Hauptursache für HABs beschreibt, zeigen Beispiele internationaler Studien, dass temperaturbedingte Anomalien eher den Zeitpunkt und die Intensität der Blütenentwicklung beeinflussen können, anstatt als direkter Auslöser zu fungieren. Diese Arbeit konzentriert sich auf die Beziehung zwischen MHW-Ereignissen und HABs in der Oberflächenschicht der Ostsee unter Verwendung von Sentinel-2-Daten innerhalb der cloudbasierten Plattform Google Earth Engine (GEE). Ziel ist es, die Korrelation und den Einfluss zwischen MHWs und HABs zu untersuchen und Zeiträume zu identifizieren, in denen sich beide überschneiden. Dafür wird die Intensität der Cyanobakterienblüte durch Berechnung eines Phycocyanin-basierten Index quantifiziert, während MHW-Ereignisse aus Aufzeichnungen der Meeresoberflächentemperatur (SST) unter Verwendung eines schwellenwertbasierten Detektionsalgorithmus identifiziert werden. Die Ergebnisse zeigen keinen signifikanten Zusammenhang zwischen marinen Hitzewellen und der Häufigkeit des Auftretens von Cyanobakterienblüten. Es wurde jedoch eine signifikante Korrelation zwischen der Dauer der MHW und der Intensität der Blüte beobachtet. Diese Ergebnisse legen nahe, dass marine Hitzewellen eher als verstärkender Faktor für bereits etablierte Blütenbedingungen wirken, als dass sie als primärer Auslöser dienen. Diese Erkenntnisse tragen zur folgenden Diskussion darüber bei, ob MHWs tatsächlich als nennenswerter Treiber von Cyanobakterienblüten in der Ostsee betrachtet werden könnten, gestützt durch bereits vorhandene Studien, Forschungsergebnisse und Literatur.

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II. List of abbreviations

HAB	- harmful algal bloom
MHW	- marine heatwave
PC	- Phycocyanin-concentration
GEE	- Google Earth Engine
JJA	- June, July, August
SST	- Sea surface temperature

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1. Introduction

Marine Heatwave Events have been growing in frequency and intensity in most regions of the world (Carstensen et al. 2014; Oliver et al. 2018; IPCC AR6 2023). The trends of MHW-Events in stratified marginal seas like the Baltic Sea have been studied extensively (Gröger et al. 2024, Safonova et al. 2024). At the same time an increase in Harmful Algal Blooms (HAB's) was described by multiple sources for this region (HELCOM 2023, Olofsson et al. 2020). While most studies describe increased nutrient input as the main driver of HAB's (Löptien & Dietze 2022), examples from global studysites have shown that these events are sometimes accompanied, and in some cases even thought to be driven by Marine Heatwave Events (Roberts et al. 2019). In this study remote sensing is used to investigate the correlation between surface MHW-Events and harmful cyanobacteria blooms in the Baltic Sea. By creating indexes described by Akbarnejad Nesheli et al. (2024) it is possible to distinguish between general phytoplankton-blooms and harmful cyanobacteria-blooms. The Results of this step are compared to known MHW-Events and a correlation-index is created between the two events. Finally, the findings are evaluated in relation to current scientific literature to determine whether MHW-events can be considered a significant driver of HABs in the Baltic Sea.

2. Thematic background

2.1 Marine heatwaves

Marine heatwaves (MHWs) are prolonged periods of anomalously high SST that occur in marine environments and can have significant ecological and economic consequences (Dabulevičienė and Servaitė 2024). These events represent extreme ocean temperature anomalies and will occur more frequently and intensely in the present and future due to the results of the climate change (Marin et al. 2021, IPCC 2021).

A scientific accepted framework for identifying MHWs was proposed by Hobday et al. (2016). According to their definition, a MHW occurs when SST exceeds a seasonally varying climatological threshold, typically defined as the 90th percentile of the historical temperature distribution, for at least five consecutive days (Figure 1, Hobday et al. 2016).

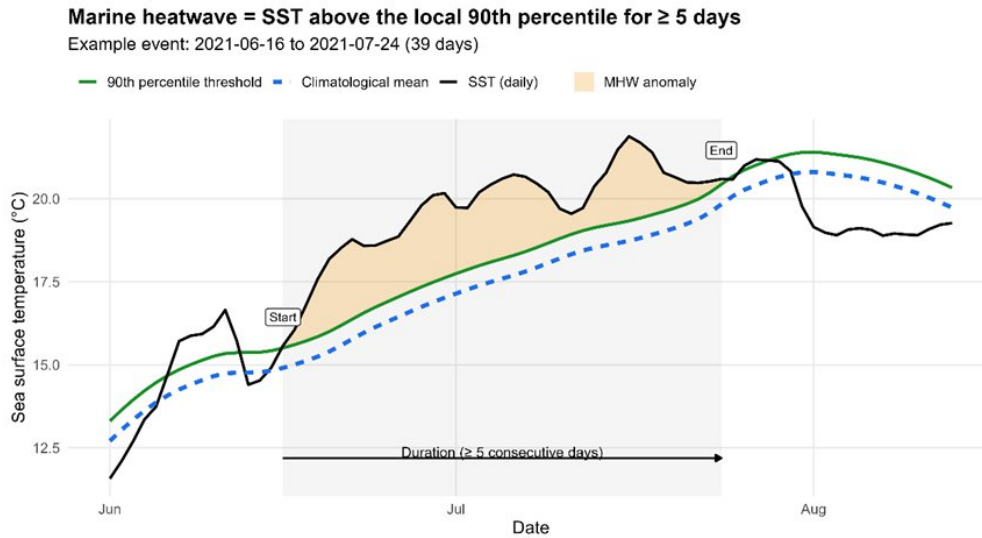


Figure 1: Visualization of the definition of MHWs in R. The Green Line represents the 90th Percentile threshold of the climatological mean baseline, represented by the dotted blue line. The black line represents the daily SST from our dataset. When the black line is higher than the green line for 5 or more days, it is conducted as MHW (Own depiction; Data: NOAA OISST v2.1; Definition by Hobday et al. 2016).

The percentile-based approach allows the identification of temperature extremes relative to local climatic conditions. Therefore, MHWs are considered as statistical temperature extremes, meaning that they represent unusually warm conditions compared to the seasonal variability at a specific location (Capotondi et al. 2024). MHWs can vary in duration from several days to months and may cover large spatial areas (Dabulevičienė and Servaitė 2024). Their ecological impacts are diverse and include shifts in species distributions, altered ecosystem functioning or increased mortality of marine organisms (Dabulevičienė and Servaitė 2024). They are also linked to the occurrence of HABs (Roberts et al. 2013; Dabulevičienė and Servaitė 2024).

2.2 Harmful Algal Blooms

HABs refer to rapid increases in the abundance of algae or cyanobacteria that can negatively affect aquatic ecosystems, human health, and economic activities (Nesheli et al. 2024; Roberts et al. 2013, Smayda 2003). Algal blooms are a natural component of marine ecosystems and considered harmful when they produce toxins or generate excessive biomass of which their rapid proliferation can result in severe environmental consequences (Lo et al. 2026). Certain algae and cyanobacteria produce biotoxins that can accumulate in marine organisms and pose risks to wildlife, livestock and humans (Bouda et al. 2025). In addition, large accumulations of algal biomass can lead to oxygen depletion, known as hypoxia, during decomposition processes which may result in fish kills and ecosystem degradation (Lo et al. 2026; Nesheli et al. 2024).

In the Baltic Sea, HABs are commonly associated with cyanobacteria blooms, which frequently occur during the summer months under favourable environmental conditions such as high-water temperatures, stable stratification and sufficient nutrient availability (IOW 2021; Telesh et al. 2025). Cyanobacterial blooms lead to eutrophication of the Baltic Sea, due to their high amount of biomass (Telesh et al. 2025, Zhou et al. 2022). Because of complex plankton dynamics, that depend on multiple interrelated and highly changeable external abiotic triggers, these events are still hard to predict (Telesh et al. 2025, Fernandes-Salvador et al. 2021).

Several studies have suggested that environmental factors such as increasing SST may enhance the development and persistence of cyanobacterial blooms (HELCOM 2023; Olofsson et al. 2020; Roberts et al. 2019). Warmer water temperatures can stimulate cyanobacterial growth rates and promote water column stratification which reduces vertical mixing and creates favorable conditions for HABs (Telesh et al. 2025, Zhou et al. 2022). Consequently, climate-related changes in the SST may influence the frequency and intensity of HAB events in the Baltic Sea.

2.3 Linking Marine Heatwaves and Cyanobacteria Blooms

Recent research suggests that MHWs may influence the development and intensity of HABs by increasing water temperatures, which can enhance HAB growth and promote bloom persistence (HELCOM 2023; Olofsson et al. 2020; Roberts et al. 2019). However, the potential relationship between MHWs and HABs for the Baltic Sea remains insufficiently explored, particularly using satellite-based observations. This study therefore investigates whether MHW may act as a potential driver and increase the frequency or intensity of HAB events.

3. Material and Methods

The methodological structure of this work is based on satellite data, which was processed and statistically analyzed using the remote based Platform Google Earth Engine (GEE) and R.

3.1 Study Area

This study focuses on the Baltic Sea, a semi-enclosed marginal sea in Northern Europe characterized by strong seasonal stratification, relatively low salinity, and pronounced spatial and temporal variability in physical and biogeochemical conditions (Voss et al. 2025, Fig. 2).

The region of interest covered the central Baltic Sea basin and adjacent sub-basins relevant for cyanobacterial bloom formation. Our selected study area is defined as a polygon shapefile from IHO in the GEE and used consistently for all subsequent analyses (Marine Regions 2026, Fig. 2).

The temporal focus of this study is restricted to the summer months June, July and august (JJA), as cyanobacterial blooms in the Baltic Sea mostly appear during this period (IOW 2021; Telesh et al. 2025).

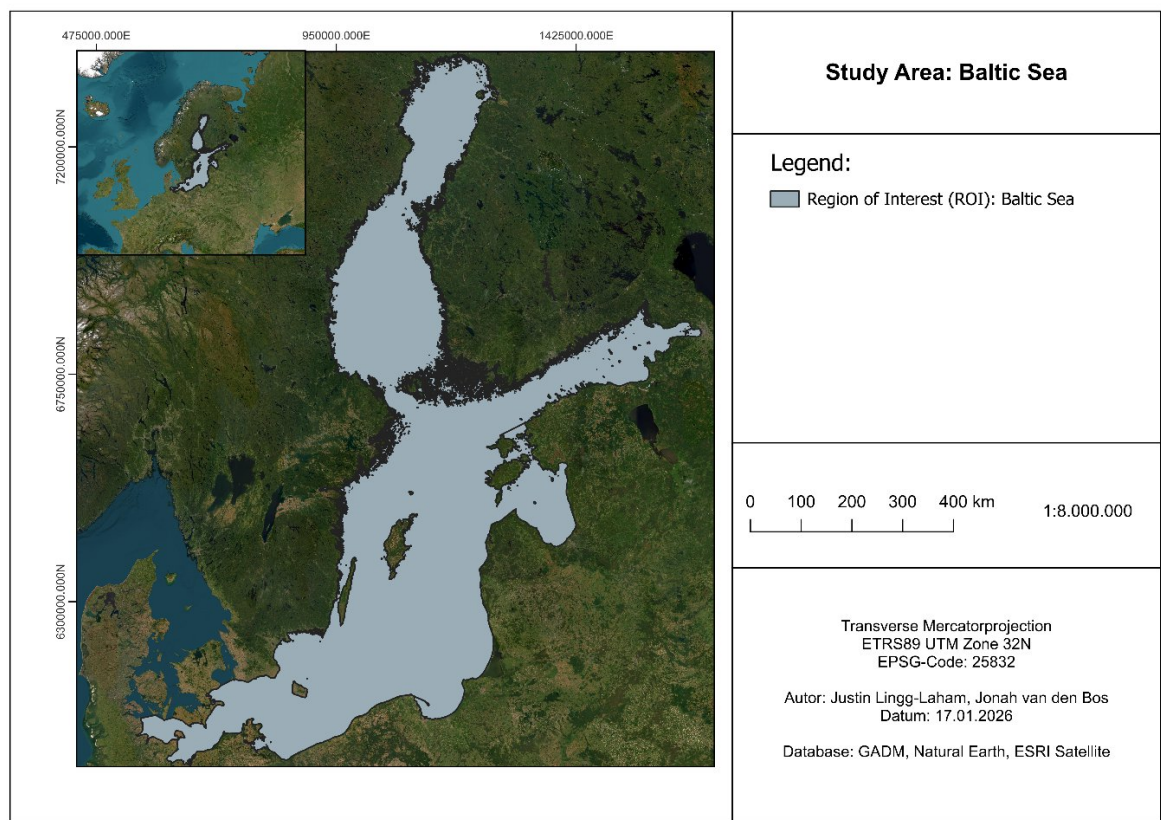


Figure 2: Map of the Region of Interest. The Baltic Sea (own Depiction).

3.2 Data

Two satellite derived datasets were used in this study: Sentinel-2 Level-2A surface Reflectance (SR) for cyanobacterial bloom detection and NOAA OISST v2.1 for MHW detection (Tab. 1).

Table 1: Summary of all Datasets used in this study (Own Depiction).

Productname	Collection Snippet	Dataset Availability	Spatial Resolution	Reference
Sentinel-2 Level-2A surface reflectance imagery (SR)	ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")	2015-Present	10 meter	Google Earth Engine: https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel2.html#sentinel-2-level-2a-top-of-canopy-toc
NOAA OISST v2.1	ee.ImageCollection("NOAA/CDR/OISST/V2_1")	1981-Present	27.830 meter	Google Earth Engine: NOAA's National Centers for Environmental Information (http://www.ncei.noaa.gov)
Marine Regions: IHO Sea Area	roi	2021	-	Marineregions.org

To assess bloom intensity, imagery from the COPERNICUS/S2_SR_HARMONIZED image collection was processed in GEE. This dataset provides atmospherically corrected surface reflectance data from Sentinel-2 and includes multispectral optical information suitable for aquatic remote sensing applications (Sentinel 2)

Only images intersecting the ROI were considered. To reduce atmospheric contamination, scenes were filtered by a maximum CLOUDY_PIXEL_PERCENTAGE of 30%. Additional cloud and shadow masking was applied using both the QA60 cloud mask and the Scene Classification Layer (SCL) (Wright et al. 2024). Only water pixels were retained in the analysis, while pixels classified as cloud, cirrus, or cloud shadow were excluded (Wright et al. 2024).

MHW conditions were derived from the NOAA/CDR/OISST/V2_1 daily sea surface temperature product. This dataset provides daily SST at approximately 0.25° spatial resolution and is widely used in MHW studies (Duerr et al. 2011).

The OISST dataset was spatially filtered to the buffered ROI and converted from scaled integer values to degrees Celsius by multiplication with 0.01. The MHW analysis included all daily summer observations from 2015 to 2025 (Duerr et al. 2011). In addition, a climatological baseline period from 1991 to 2020 was used to calculate daily threshold values for MHW detection.

3.3 Methods

The methodological workflow consists of three main steps: (1) detection of cyanobacterial bloom intensity from Sentinel-2 imagery, (2) derivation of MHW conditions from daily SST data and (3) statistical comparison of both variables in R (Fig. 3)

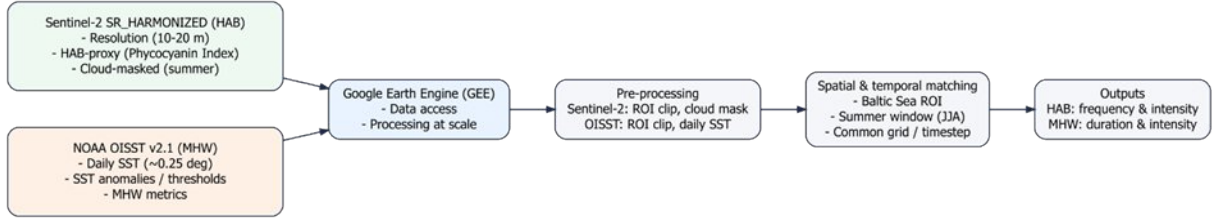


Figure 3: Simple Flowchart of the Analysis-Workflow (Own Depiction).

3.3.1 Phycocyanin proxy calculation

To estimate cyanobacterial bloom intensity, a phycocyanin-based proxy was calculated for each Sentinel-2 image following the approach proposed by Akbarnejad Nesheli et al. (2024). The method is based on the spectral response of algal biomass in the red-edge region and uses Sentinel-2 bands B4 (665 nm), B5 (705 nm), and B8A (865 nm).

In the beginning, reflectance values were scaled by multiplying the values by 0.0001. A linear spectral baseline at 705 nm was calculated between B4 and B8A. The maximum peak height was derived as:

$$MPH = R_{705} - baseline_{705}$$

where R_{705} is the reflectance in band B5 and $baseline_{705}$ is the linearly interpolated baseline reflectance at 705 nm (Nesheli et al. 2024). The maximum peak height signal was then converted to estimated phycocyanin concentration (PC) using the empirical one-day model of Akbarnejad Nesheli et al. (2024) Table 8:

$$PC_{\mu g/L} = 3047.6 \times MPH + 0.626$$

Negative values were truncated to zero. It is important to note that this variable does not represent a direct in situ measurement of phycocyanin, but rather a modelled surface proxy for cyanobacterial bloom intensity derived from satellite data (Fig. 4).

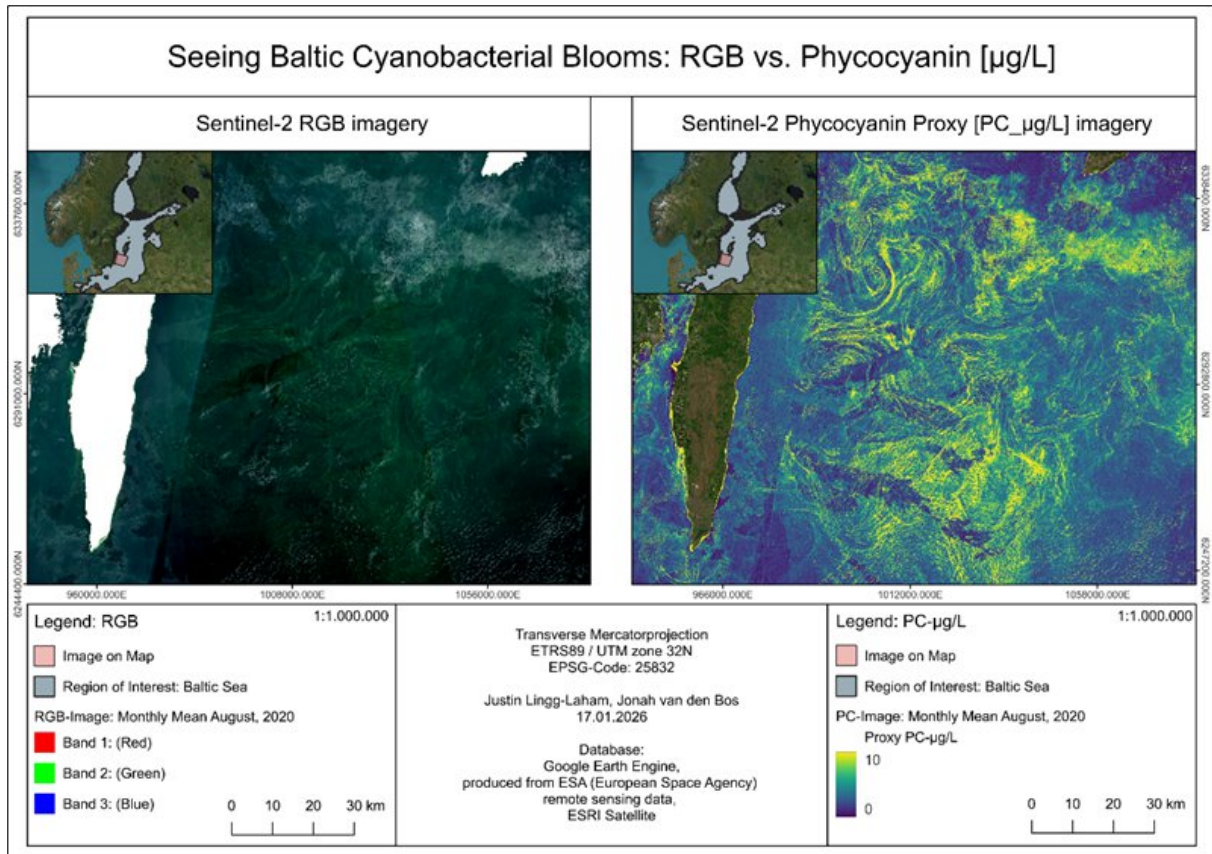


Figure 4: Map of HABs event based on the calculated phycocyanin proxy. Left an RGB image of a Snipped of the ROI, Right the calculated phycocyanin-proxy in $\mu\text{g/L}$ for comparison (Own Depiction).

3.3.2 Monthly bloom metrics

For each summer month and year, the image collection was aggregated over the ROI and summarized into monthly bloom metrics. The following variables were calculated:

- **Image_count:** Nuber of valid Sentinel-2 scenes used
- **PC_mean:** spatial mean of the monthly mean phycocyanin proxy
- **PC_median:** spatial mean of the monthly median phycocyanin proxy
- **PC_max:** maximum phycocyanin proxy value within the ROI
- **PC_p90:** 90th percentile of the phycocyanin proxy, used as an indicator of bloom intensity
- **bloom_fraction:** fraction of pixels or observations exceeding a bloom threshold
- **bloom_count_approx:** approximate bloom count, calculated as **bloom_fraction** and multiplied by **image_count**

Following Akbarnejad Nesheli et al. (2024), a fixed threshold of 10 $\mu\text{g/L}$ was used to identify moderate bloom conditions (Fig. 4). All monthly results were exported from GEE as CSV files for further processing in R.

3.3.3 Marine heatwave detection

To detect MHWs the conceptual framework of Hobday et al. (2016) was implemented. Daily SST values from NOAA OISST were compared to a climatological threshold derived from the baseline period 1991-2020 (Hobday et al 2016).

For each day of year during summer months, the 90th percentile SST threshold was calculated from the baseline time series. This produced a seasonally varying threshold that reflects local climatological conditions rather than an absolute temperature value, because MHW are statistical temperature extremes (CRAN-R 2026, Dabulevičienė and Servaitė 2024).

For each daily summer observation between 2015 and 2025, the corresponding day of year specific threshold was assigned. The daily SST and threshold values were then spatially averaged across the ROI and exported as a daily time series (CRAN-R 2026, Dabulevičienė and Servaitė 2024).

The actual event detection was performed in R using the package `heatwaveR`. Daily SST values (“temp”) and threshold values (“thresh”) were reformatted to match the requirements of the package (CRAN-R 2026, Schlegel 2026). MHW-Events were defined as periods of at least five consecutive days with SST above the threshold (Hobday et al. 2016). From the detected events the following metrics were calculated monthly and annual: event duration, maximum intensity, cumulative intensity (CRAN-R 2026, Schlegel 2026).

3.3.4 Statistical Analysis

To investigate the relationship between MHWs and cyanobacterial blooms, the daily MHW information was aggregated to the monthly scale to match the temporal resolution of the phycocyanin dataset.

Two main analyses were conducted in this work:

1. *Comparison of bloom intensity in months with and without MHW conditions*

Monthly phycocyanin values were grouped into “MHW month” and “Without MHW” based on whether any MHW-days occurred within the corresponding month. Differences in bloom intensity between these groups were visualized using boxplots.

2. *Correlation between MHW duration and bloom intensity*

For each summer month, the total number of MHW days was calculated and matched with the monthly phycocyanin proxy metrics. The relationship between MHW duration and bloom intensity was then assumed using Spearman's rank correlation, as the data were not assumed to be normally distributed and the analysis focused on monotonic rather than strictly linear relationships.

The detailed workflow can also be found in the following GitHub-Repository containing results, code and figures:

<https://github.com/geojussl/Marine-Heatwaves-An-overlooked-Driver-of-Baltic-Cyanobacteria-Blooms-.git>



Figure 5: QR-Code for the GitHub-Repository (Own Depiction).

4. Results

The monthly mean phycocyanin-index was slightly higher during summer months that are within or contained a MHW-Event compared to months without MHW-Events (Figure 6). However, the ranges of both groups overlapped considerably. Multiple high outliers were observed in months without MHW-Events, showing greater variability in PC under non-MHW conditions. Despite these extreme values, the values were lower than in months with MHW-Events. No clear relationship between the presence of MHW conditions and the monthly mean phycocyanin index can be deduced from this comparison.

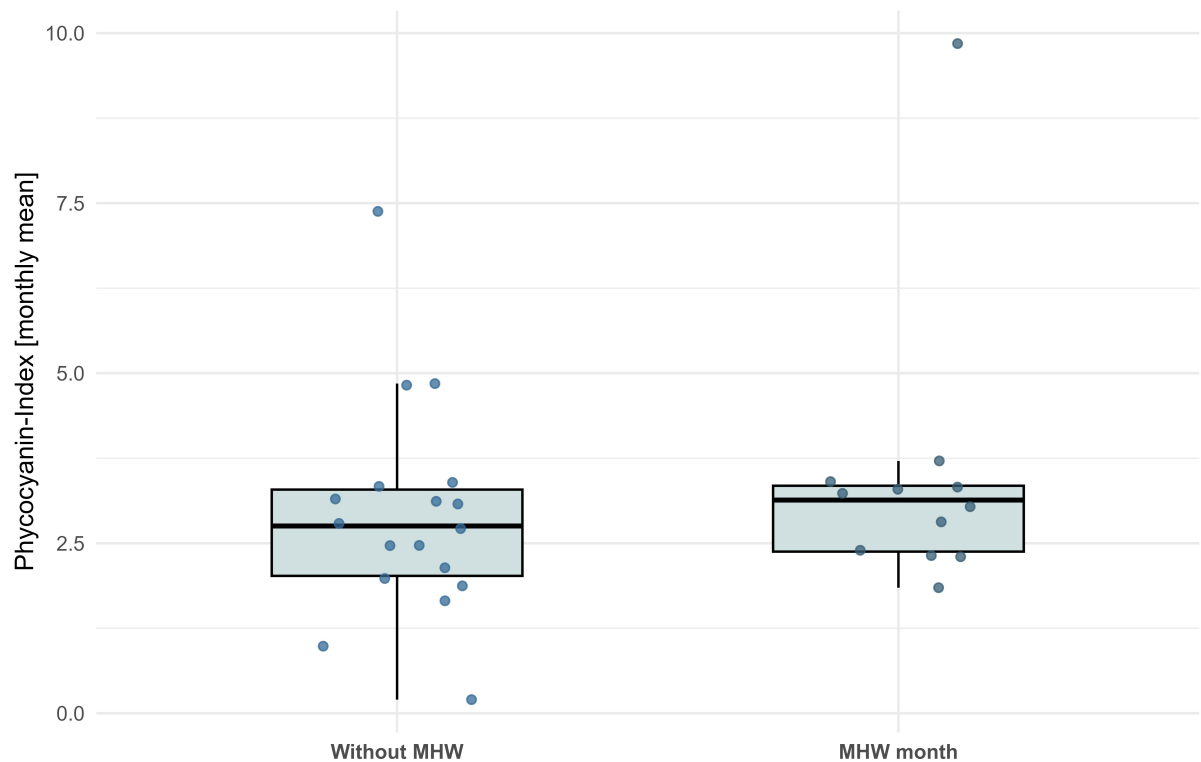


Figure 6: Phycocyanin-Index during summermonths (JJA) in the Baltic Sea with and without MHWs (Own Depiction).

Peak PC during the summer months JJA in the study period of 2015–2025 increased with the duration of MHW-Events (Figure 7). Months without MHW-Events included lower peak phycocyanin values. Linear regression showed a moderate significant correlation between MHW duration and peak PC ($p < 0.05$, $n = 30$). Higher variability and outliers were observed in months without MHW-Events, while months with longer MHW duration tend to show higher phycocyanin concentrations.

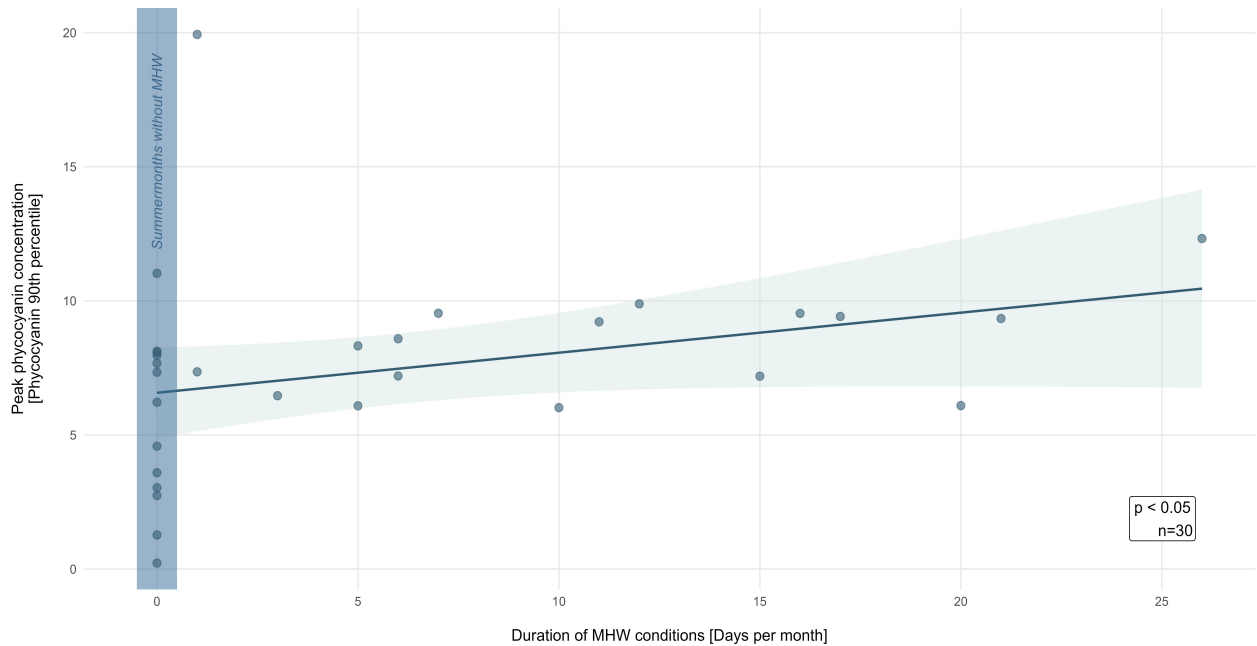


Figure 7: Correlation between MHW duration and cyanobacterial bloom intensity during summermonths (JJA) in the research period (2015-2025) (Own Depiction).

5. Discussion

While our results align with the scientific consensus for MHWs acting as a notable driver for cyanobacteria blooms in the Baltic Sea (Lips & Lips 2008, Kahru et al. 2007), they could not confirm that MHWs increase the frequency of them, as others have suggested (Chorus et al. 2021, Kahru et al. 2020).

Kahru et al. (2020) found that water temperature has an effect on the frequency of cyanobacteria accumulations in the Baltic Sea, but they also stated that the correlation between satellite-detected SST-temperature was considerably lower than the correlation between in-water measured temperatures and the frequency of blooms. This raises the question if Satellite-SST measuring procedures can give precise enough data to accurately calculate a correlation in this case. Further limitations of the methods of this study could be found in the inability to accurately represent regional temperature changes and cyanobacteria blooms, thus creating error when MHWs do not engulf the full extend of the Baltic Sea. Due to the nature of the methodology of this work, MHW-Events that do not affect the entire summermonth they appear in or the entire Baltic Sea region are still counted as such. It is possible that the low correlation between the frequency of cyanobacteria blooms and MHWs is also the result of a different counter-driver that was not analysed in this work. Other studies have suggested warm temperatures can only be a cause of blooms if all other parameters such as nutrients and/or water currents are in the optimal range (Chorus et al. 2021).

6. Conclusion

This study could not demonstrate any significant indicator that MHWs are causing cyanobacteria blooms to appear more frequently. To better understand MHW effects on HAB frequency in the Baltic Sea, further and more precise studies are required, especially those that focus on temperature data directly recorded in situ. Further research is also needed on the position of MHWs within a network of environmental parameters promoting cyanobacteria blooms in the Baltic Sea. Understanding these could help us create warning systems for, mitigating or even preventing HABs in the future.

This study did however, find evidence that MHWs act as a notable driver for HABs in the Baltic Sea. With these results, it joins a wide array of already existing studies and literature that suggest MHWs as important drivers or even catalysts of already occurring HABs. This work gives further concrete evidence that the driving property of MHWs on HABs can still be observed within modern data. This data subsequently supports the approach that more research and monitoring on MHWs and HABs are needed to protect and warn coastal populations and tourism industries but also to prevent environmental catastrophes, fish kills, biodiversity loss, and in the special case of the Baltic Sea, the creation of hypoxia driven dead zones.

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